

■# PAPER<i>353 — Double-Exponential Vacuum Decay Rate: ρ</i>SCm/ρ</i>UA Ratio with Near-Threshold Behavior

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Session: 96

■# PAPER353 — *Double-Exponential Vacuum Decay Rate: ρ_{SCm}/ρ_{UA} Ratio with Near-Threshold Behavior*

Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gokshare31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF double-exponential vacuum decay rate with near-threshold ($\pi - t \rightarrow 0$) behavior **Author:** Daniel T. Murphy

<!-- UQFF constants: $\kappa = 5.0e-4 \text{ day}^{-1}$, $[SSq] = 0.57$, $M_{UQFF} = 1.43e1 \text{ TeV}$ -->

Abstract

A novel double-exponential decay rate formula is derived for UQFF vacuum energy channels, incorporating a near-threshold correction for the approach of $t \rightarrow \pi$ (the natural oscillation phase singularity). The rate is: $\text{Rate} = (\rho_{SCm}/\rho_{UA}) \cdot \exp(-[SSq] \cdot n/26 \cdot \exp(-(\pi - t)))$. The *rho*ratio = $\rho_{SCm}/\rho_{UA} = 0.1$ establishes a 10:1 aether-to-superconductive vacuum density contrast, and the double-exponential structure ensures the rate approaches zero continuously as $t \rightarrow \pi$ (without a singularity).

2. Core Physics

2.1 Double-Exponential Decay Rate

$$\mathrm{Rate}(n, t) = \frac{\rho_{SCm}}{\rho_{UA}} \cdot \exp\left(-\frac{[SSq]}{n/26} \cdot \exp(-(\pi - t))\right)$$

2.2 Near-Threshold Behavior

As $t \rightarrow \pi$ (oscillation phase approaches singularity):

$$\exp(-(\pi - t)) \rightarrow \exp(0) = 1$$

$$\mathrm{Rate}_{\mathrm{threshold}} = \frac{\rho_{SCm}}{\rho_{UA}} \cdot \exp\left(-\frac{[SSq]}{n/26}\right) = 0.1 \cdot e^{-0.57 \cdot n/26}$$

This is the standard $[SSq]$ -compressed decay without the t -modulation — confirming the double-exponential reduces to single- $[SSq]$ at threshold.

2.3 Standard (Away From Threshold)

For $t \ll \pi$:

$$\exp(-(\pi - t)) \rightarrow e^{-\pi} \approx 0.0432$$

$$\mathrm{Rate}_{\mathrm{standard}} = 0.1 \cdot \exp\left(-\frac{0.57 \cdot n}{26}\right) \times 0.0432 = 0.1 \cdot \exp\left(-\frac{0.0247 \cdot n}{26}\right)$$

Very slow decay — most channels maintain near-full rate far from threshold.

2.4 Vacuum Density Contrast

$$\rho_{\text{ratio}} = \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} = 0.1$$

A factor-of-10 suppression: the superconductive vacuum is 10× less dense than the UA aether background, calibrated by comparing UQFF predictions to dark energy measurements (ρ_{Λ} from Planck 2018).

3. Key Values

Quantity	Formula	Value
ρ_{ratio}	$\rho_{\text{SCm}}/\rho_{\text{UA}}$	0.1
[SSq]	Canonical	0.57
Rate (threshold)	$\rho_{\text{ratio}} \cdot \exp(-[\text{SSq}]n/26)$	$0.1 \cdot e^{-(0.022n)}$
Rate (max, n=1)	Near-threshold n=1	~0.0976
Rate (far, n=26)	Away from threshold	~0.0566
Singularity point	$t = \pi$	Phase singularity

4. Physical Significance

The double-exponential form avoids a discontinuity at the phase singularity $t = \pi$ that would appear in a naive single-exponential decay. This is physically important for UQFF oscillation cycles: real vacuum oscillations do not diverge as t approaches the half-period. The outer $\exp(-(\pi-t))$ factor acts as a "near-threshold regulator," ensuring continuous differentiability of $\text{Rate}(n,t)$ at all $t \in [0, \pi]$.

The $\rho_{\text{ratio}} = 0.1$ is a key calibration: it constrains the ratio of superconductive to aether vacuum energy density, providing an observational connection to the measured dark energy density parameter ($\Omega_{\Lambda} = 0.678$ from Planck 2018).

5. Deduplication Note

****vs. PAPER_353 vs. standard [SSq] decay:**** Standard single-exponential $\exp(-[\text{SSq}]n/26)$ appears in dozens of UQFF papers; the double-exponential form with near-threshold behavior is new in PAPER_353.

****vs. PAPER_341 (calibration):**** PAPER_341 derives κ , H_{SCm} , U_{UA} values; PAPER_353 uses the $\rho_{\text{ratio}} = 0.1$ calibration separately.

6. Classification

Physics Territory: FIRST UQFF double-exponential vacuum decay rate with near-threshold regulator **Scale:** Vacuum field (universal) **CP Implementation:** DecayRateVacuumRhoRatioDoubleExpCalculator (CondensedPhysics3.py, Session 96)